

Class 9

Real-time simulations with MPS

$$\boxed{i\partial_t |\psi(t)\rangle = \hat{H} |\psi(t)\rangle} \Rightarrow |\psi(t)\rangle = e^{-i\hat{H}t} |\psi(0)\rangle$$

$$|\psi(t)\rangle \Leftrightarrow \bigcirc - \bigcirc - \bigcirc - \bigcirc - \bigcirc - \bigcirc$$

We want to evolve $\psi(t)$ while keeping the MPS form.

1D chains: the Time Evolving Block Decimation algorithm

Example: XXZ-model

$$H = \sum_i \underbrace{J(\sigma_i^x \sigma_{i+1}^x + \sigma_i^y \sigma_{i+1}^y + \Delta \sigma_i^z \sigma_{i+1}^z)}_{h_i} = \sum_i h_i$$

IMPORTANT: only nearest neighbor couplings

$$\hookrightarrow H = H_{\text{even}} + H_{\text{odd}} = \sum_k h_{2k} + \sum_k h_{2k+1}$$

Only nearest neighbor couplings
 $\Downarrow$

$$h_i = \bigcirc \otimes \bigcirc \otimes \dots \otimes \bigcirc \otimes h_i \otimes \bigcirc \otimes \dots$$

$\uparrow$
site $i, i+1$

$$[h_{2k}, h_{2\ell}] = \emptyset \quad \forall k, \ell$$

$$[h_{2k}, h_{2\ell+1}] \neq \emptyset$$

$$[h_{2k+1}, h_{2\ell+1}] = \emptyset \quad \forall k, \ell$$

$\Downarrow$

$$\left. \begin{aligned} e^{-iH_{\text{even}}t} &= \prod_k e^{-ih_{2k}t} \\ e^{-iH_{\text{odd}}t} &= \prod_k e^{-ih_{2k+1}t} \end{aligned} \right\} \text{Products of 2-site unitaries}$$

The Trotter-Suzuki approximation (1st order)

$$e^{-iH\Delta t} = e^{-iH_{\text{even}}\Delta t} e^{-iH_{\text{odd}}\Delta t} + O(\Delta t^2)$$

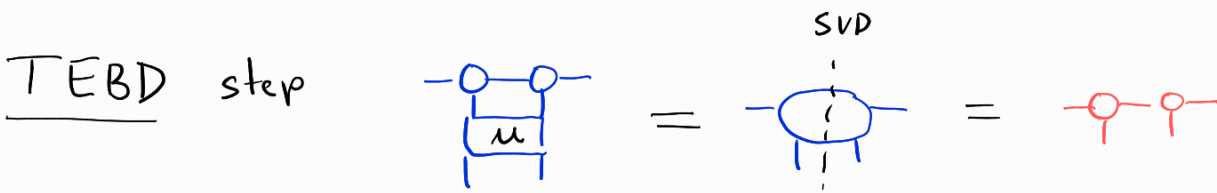
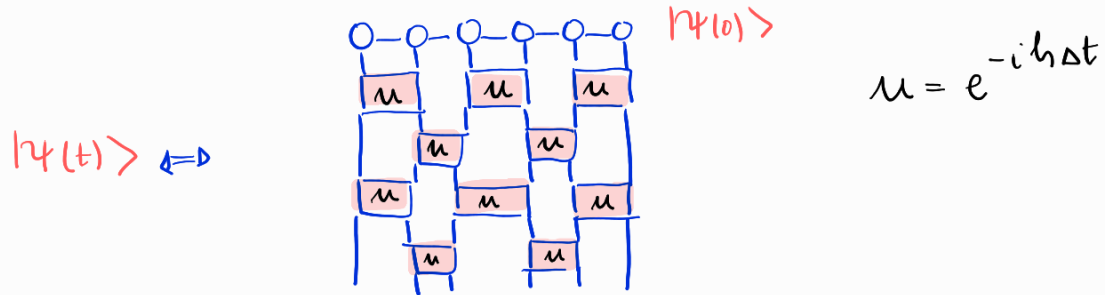
Proof: Taylor's series in Δt

$$\begin{aligned} & (1 - iH\Delta t - \frac{1}{2}H^2\Delta t^2 + \dots) - (1 - iH_{\text{even}}\Delta t - \frac{1}{2}H_{\text{even}}^2\Delta t^2 + \dots)(1 - iH_{\text{odd}}\Delta t - \frac{1}{2}H_{\text{odd}}^2\Delta t^2 + \dots) = \\ & \underbrace{(1-1)}_{\emptyset} - i\Delta t \cdot \underbrace{(H - H_{\text{even}} - H_{\text{odd}})}_{\emptyset} - \frac{\Delta t^2}{2} \cdot \underbrace{(H^2 - H_{\text{even}}^2 - H_{\text{odd}}^2 - 2H_{\text{even}}H_{\text{odd}})}_{[H_{\text{odd}}, H_{\text{even}}]} + \dots \end{aligned}$$

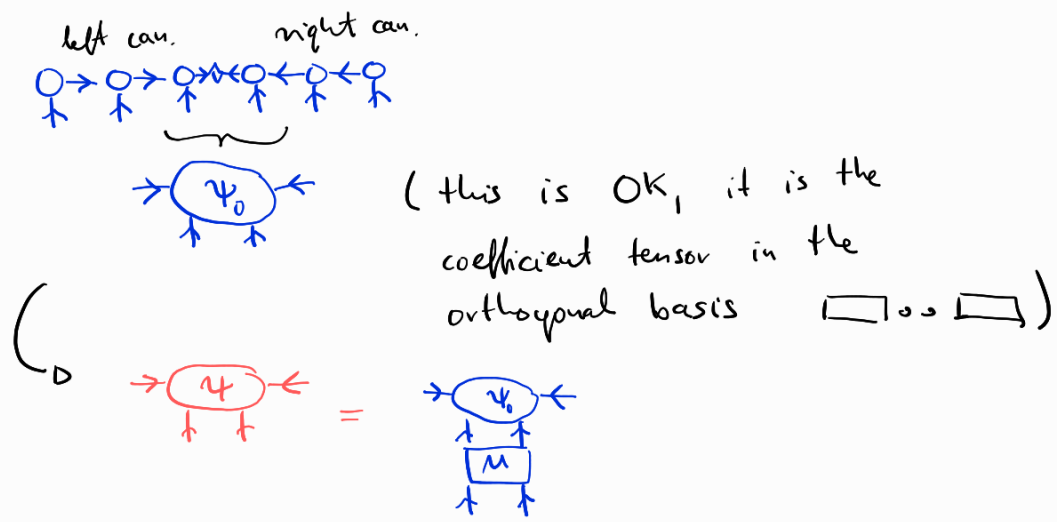
$\Downarrow$

$$e^{-iHt} = (e^{-iH_{\text{even}}\Delta t} e^{-iH_{\text{odd}}\Delta t})^{\frac{t}{\Delta t}} + O(\underbrace{\frac{t}{\Delta t} \cdot \Delta t^2}_{t \cdot \Delta t} \text{ Trotter Error})$$

$\Downarrow$



issue: Schmidt decomposition requires orthogonal basis



Now we can perform SVD on Y

ψ
 D

↳ We have to move the orthogonality center.

↳ or can we avoid it?

C_D Vidal's form of MPS

Left-canonical matrices

$$A^{(1)} = \Gamma^{(1)}$$

$$A^{(i)} = \lambda^{(i-1)} \Gamma^{(i)} \rightarrow \text{diamond} \leftarrow \text{circle} \rightarrow$$

Right-canonical

$$B^{(L)} = \Gamma^{(L)}$$

$$B^{(i)} = \Gamma^{(i)} \lambda^{(i)}$$

2-site tensor

The diagrammatic equation shows the two-point function of the composite operator Ψ (represented by a circle with Ψ inside) is equal to the sum of two diagrams. The first diagram on the right is a chain of four vertices: a diamond, a circle, a diamond, and a circle, connected by horizontal lines. The second diagram is identical to the first but with the two outer diamond vertices crossed out.

TEBD Algorithm (1 microstep)

(i) =

ii)

$\psi(i)$ = $\psi(i)_0$ with legs connected to u

(ii)

$\tilde{A}^{(i)} \quad \tilde{\lambda}^{(i)} \quad \tilde{B}^{(i+1)}$

$\chi^{(i+1)}$

SVD

$$i v) \quad \begin{array}{c} \tilde{\Gamma}^{(i)} \\ \leftarrow \bigcirc \rightarrow \\ \uparrow \end{array} = \frac{\lambda}{\lambda^{(i-1)}} \begin{array}{c} \tilde{A}^{(i)} \\ \leftarrow \diamond \rightarrow \bigcirc \rightarrow \\ \uparrow \end{array}$$

$$\hat{\lambda}^{(i-1)} = \lambda^{(i-1)}$$

$$\hat{\lambda}^{(i+1)} = \lambda^{(i+1)}$$

They are
unchanged.

(u is unitary!)

Numerical issue : $1/\lambda$ accuracy

- With the microscopic step, we can update the MPS for every 2-site gates

- infinite - chain variant:

initial state with (periodic) translational invariance

e.g. $|\psi_0\rangle = \dots \uparrow \downarrow \uparrow \downarrow \dots$

$$\begin{aligned} \Gamma_{11}^{(A)\uparrow} &= 1 & \lambda_1^{(A)} &= 1 & \Gamma_{11}^{(B)\uparrow} &= 0 & \lambda_1^{(B)} &= 1 \\ \Gamma_{11}^{(A)\downarrow} &= 0 & & & \Gamma_{11}^{(B)\downarrow} &= 1 & & \end{aligned}$$

$$|\psi_0\rangle = \dots \Gamma^{(A)} \lambda^{(A)} \Gamma^{(B)} \lambda^{(B)} \Gamma^{(A)} \lambda^{(A)} \Gamma^{(B)} \lambda^{(B)} \dots$$

↳ the MPS chain has no ends!

↳ only 2 Γ and 2 λ matrices

updates

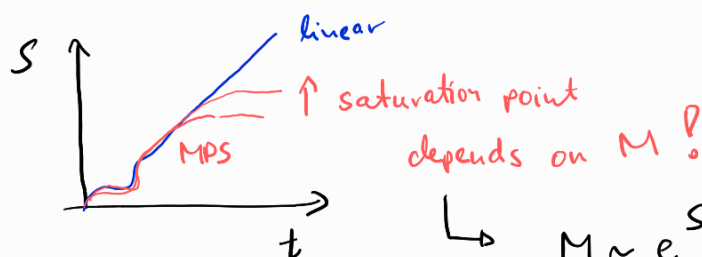
$$\left. \begin{aligned} \Gamma^A \lambda^A \Gamma^B &\leadsto \Gamma^{A'} \tilde{\lambda}^A \Gamma^{B'} \\ \Gamma^{B'} \lambda^B \Gamma^{A'} &\leadsto \tilde{\Gamma}^B \tilde{\lambda}^B \tilde{\Gamma}^A \end{aligned} \right\} \begin{pmatrix} \Gamma^A \\ \Gamma^B \\ \lambda^B \\ \lambda^A \end{pmatrix} \Rightarrow \begin{pmatrix} \tilde{\Gamma}^A \\ \tilde{\Gamma}^B \\ \tilde{\lambda}^B \\ \tilde{\lambda}^A \end{pmatrix}$$

- Entropic limitations

$$S = - \sum_{\alpha} |\lambda_{\alpha}|^2 \log |\lambda_{\alpha}|^2$$

"Global quench" $H_0 |\psi_0\rangle = E_0 |\psi_0\rangle$ (ground state)

but at $t=0$ $H_0 \rightarrow H$ (we change the parameters)

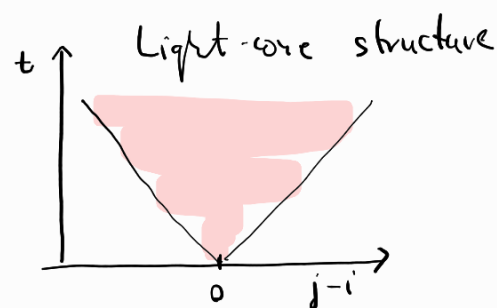


↳ $M \sim e^S$

↳ long times require exponential bond dimensions

Correlations (equal time)

$$C_{ij}^{zz}(t) = \langle \psi(t) | \sigma_i^z \sigma_j^z | \psi(t) \rangle$$

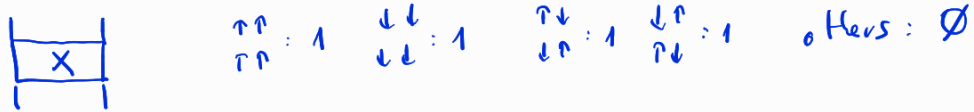


Long(er) range interactions?

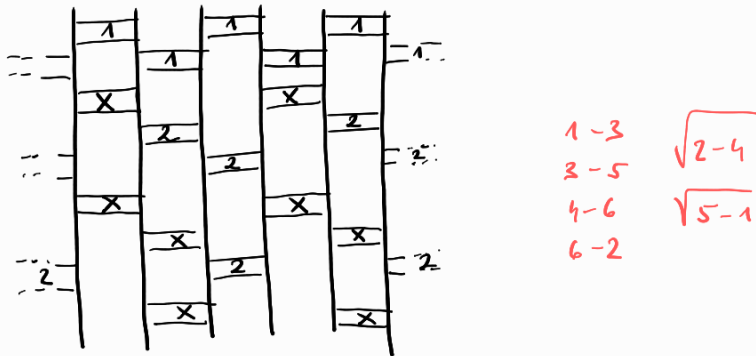
e.g. $H = \sum_i J_1 (\sigma_i^x \sigma_{i+1}^x + \sigma_i^y \sigma_{i+1}^y + \Delta \sigma_i^z \sigma_{i+1}^z) + J_2 (\sigma_i^x \sigma_{i+2}^x + \sigma_i^y \sigma_{i+2}^y + \Delta \sigma_i^z \sigma_{i+2}^z)$

TEBD needs nearest neighbor gates!

↳ we can swap second neighbors next to each other by swap gates



↳ full time step (iTEBD, unit cell: 6)



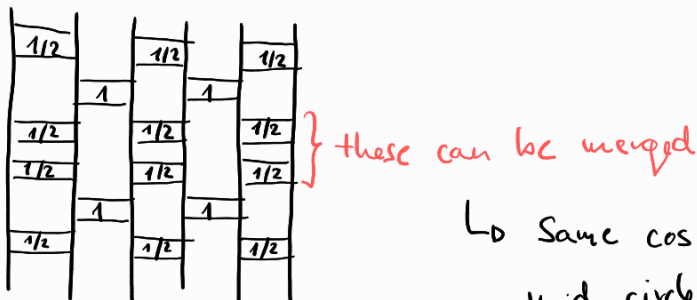
2nd order Trotter

$$e^{-iH\Delta t} = e^{-iH_{\text{Even}} \frac{\Delta t}{2}} e^{-iH_{\text{Odd}} \Delta t} e^{-iH_{\text{Even}} \frac{\Delta t}{2}} + O(\Delta t^3)$$

proof:

$$\begin{aligned} e^{-iH\Delta t} - e^{-iH_{\text{Even}} \frac{\Delta t}{2}} e^{-iH_{\text{Odd}} \Delta t} e^{-iH_{\text{Even}} \frac{\Delta t}{2}} &= \\ &= (1-1) - i\Delta t (H - H_{\text{Even}} - H_{\text{Odd}}) - \frac{\Delta t^2}{2} \left(H^2 - H_{\text{Even}}^2 \left(\frac{1}{4} + \frac{1}{4} + \frac{1}{2} \right) - \right. \\ &\quad \left. - H_{\text{Odd}}^2 - H_{\text{Even}} H_{\text{Odd}} - H_{\text{Odd}} H_{\text{Even}} \right) + \dots \end{aligned}$$

this is 0 too



↳ Same cost as 1st order, if no mid circle measurements are performed.

Tangent space methods

H : MPO, with arbitrary couplings

How to evolve the MPS?

The "Basis Update and Galerkin" (BUG) algorithm

2 subsystem case

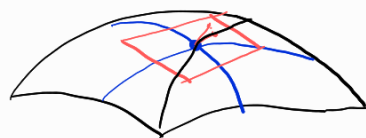
$$| \psi \rangle = \sum_{i=1}^{\dim_A} \sum_{j=1}^{\dim_B} \sum_{\alpha, \beta=1}^M U_{i\alpha} S_{\alpha\beta} V_{j\beta} |i\rangle |j\rangle$$

$\mathcal{H}_A \otimes \mathcal{H}_B$

D Unitaries

$M \ll \dim_A, \dim_B$, low rank approximation

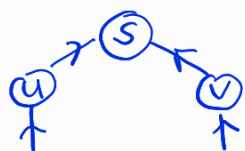
For fixed M , the accessible states form a manifold



Tangent space method:

we step in the tangent space and project back to the manifold.

$$i\partial_t |\psi_M\rangle = P_M H |\psi_M\rangle$$



U, S, V : coordinates

Naive integrator

$$i\partial_t (USV) = i\dot{U}SV + iU\dot{S}V + iUS\dot{V} = H \cdot USV$$

{ update rules

$$U(t+\Delta t) = U(t) - i \cdot \Delta t \cdot (H \cdot USV) \cdot V^{-1} \cdot S^{-1}$$

• issues:

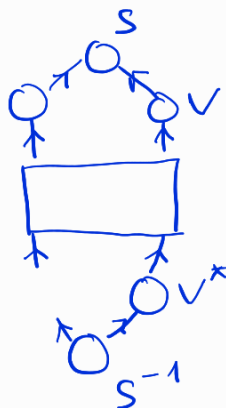
→ unitarity of U ?

→ " S^{-1} " $\Rightarrow$ division by small

singular values $\Rightarrow$ large change



High curvature



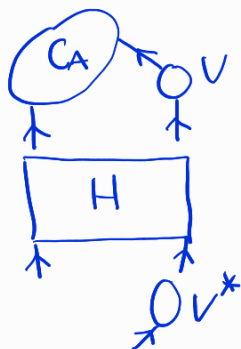
↳ solution

$$C_{i\beta}^A = \sum_{\alpha} U_{i\alpha} S_{\alpha\beta}$$

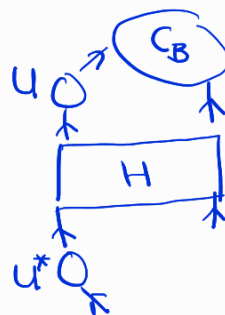
$$C_{\alpha j}^B = \sum_{\beta} S_{\alpha\beta} V_{j\beta}$$

$$i.) C^A(t+\Delta t) = C^A(t) - i\Delta t (H \cdot C^A \cdot V) \cdot V^{-1}$$

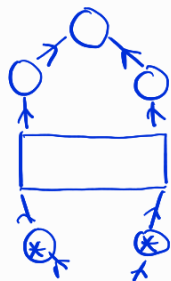
⇓
new U by QR o.



$$ii.) C^B(t+\Delta t) = C^B(t) - i\Delta t U^{\dagger} H (U C^B)$$



$$iii.) S(t+\Delta t) = S(t) - i\Delta t U^{-1} H (U S V) V^{-1}$$



$$iv.) U(t+\Delta t) = \text{"Q"} \text{ from QR decomposition of } C^A(t+\Delta t)$$

$$V(t+\Delta t) = \text{"Q"} \text{ from QR decomposition of } C^B(t+\Delta t)$$

↳ which basis to use in S-step?

$U(t), V(t)$ (explicit Euler)

$U(t+\Delta t), V(t+\Delta t) \Rightarrow$ initial norm loss!

↳ or $\left. \begin{array}{l} \text{orth } [U(t), U(t+\Delta t)] \\ \text{orth } [V(t), V(t+\Delta t)] \end{array} \right\} \begin{array}{l} \bullet \text{ Rank adaptivity} \\ \bullet \text{ increased cost} \end{array}$